

Effect of elemental interaction on microstructure and mechanical properties of FeCoNiCuAl alloys

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ABSTRACT

FeCoNiCuAlX (X refers to Si, Cr, Ti, Zr and Nd) alloys were prepared using a suck-casting method. The effect of various elements on phase constituents, microstructures and mechanical properties of the alloys was investigated using X-ray diffraction (XRD), scanning electron microscopy, and compressive tests. It has been found that the microstructure and phase constituents remain unchanged when the Si, Cr and Ti are added into the FeCoNiCuAl alloy, which have a typical cast dendrite microstructure consisting of a dominated body-centered-cubic (BCC) solid solution and a face-centered-cubic (FCC) solid solution. However, the intermetallic compounds are formed in the alloys with the addition of Zr or Nd element. The compressive strength and plasticity of the alloys are enhanced by the addition of the Si, Cr and Ti, and retarded by the addition of the Zr or Nd element. The results have been discussed in aspects of atomic size difference, electronegativity difference, valence electron concentration and the mixing enthalpy among the elements in the alloys.

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1. Introduction

For hundreds of years, traditional alloys, such as Fe-, Al-, Cu-, Mg-, Ti-, Ni-, TiAl-, NiAl- and FeAl-based alloys, are typically composed of one or two principal elements. The properties of the traditional alloys can be modified by minor additions of other elements. However, the recently developed high-entropy (HE) alloys present a novel strategy for alloy design and expand the scope of the traditional alloys [1,2]. The HE alloys typically consist of at least five principal elements with concentration between 5 at% and 35 at%, and often form one or two solid solutions with simple structures. It has been documented that the HE alloys display unique properties which are attributed to the high mixing entropy, lattice distortion, and sluggish diffusion [1]. They show great potential for engineering applications including tools, molds, and mechanical parts due to their promising properties in hardness, wear resistance, corrosion resistance, high-temperature softening resistance and oxidation resistance [1–9].

Many HE alloys have been exploited in the past decade. It has been reported that the small atomic size differences and near-zero values of the absolute enthalpy of mixing are responsible for the formation of disorder solid solutions [8]. The microstructure and properties of the alloys display strong dependence on the composition [9–22]. Vanadium addition to the $\text{Al}_{0.5}\text{CoCrCuFeNi}$ alloys enhances the formation of a body-centered-cubic (BCC)

structure and a σ phase [9]. The face-centered-cubic (FCC)-type Al promotes the transformation from FCC phase to a BCC phase, and the phase constitutions in the as-cast alloys can be adjusted by controlling the Al addition [10,11]. The addition of Ti in the CoCrCuFeNiTi_x and AlCoCrFeNiTi_x HE alloys modifies the phase constitution and properties [12,13]. Liu et al. reported that the mixing enthalpy determines whether the solid solution phases or compounds form in the nearly equiatomic multi-principal alloy systems, and the valence electron concentration is a key parameter to control the phase stability for the FCC or BCC solid solutions [21]. In our previous work, we reported that Co element has significant influence on the microstructure and mechanical properties of the $\text{FeCo}_x\text{NiCuAl}$ quinary alloys [22]. Following the Hume–Rothery rule, crystal structure, atomic size difference, electronegativity difference ($\Delta\chi$), and electron concentration of the constituent elements are key factors in determining the formation of the solid solution. In this work, Si, Cr, Ti, Zr and Nd, which have a wide range of atomic size, electronegativity and electron concentration, were added into the FeCoNiCuAl alloy to form equal-molar multi-principal alloys, and their effect on the phase constituents, microstructure and compressive properties of the FeCoNiCuAlX alloys are investigated to understand elemental interaction in the alloys.

2. Experimental details

Alloy ingots with nominal compositions of FeCoNiCuAlX (X refers to Si, Cr, Ti, Zr or Nd element in the paper) were prepared

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by arc-melting of the mixture of the constituent elements with purity better than 99.9 wt% in a water-cooled copper hearth under a titanium-gettered high-purity argon atmosphere. The ingots were remelted at least five times to assure chemical homogeneity. The weight loss after melting is 0.11%, 0.24%, 0.05%, 0.28%, 0.17%, and 3.25% for the FeCoNiCuAl, FeCoNiCuAlSi, FeCoNiCuAlCr, FeCoNiCuAlTi, FeCoNiCuAlZr, and FeCoNiCuAlNd alloys, respectively. The cylindrical rods with 5 mm in diameter and 40 mm in length were prepared using in-situ suck-casting into a copper mold under an argon atmosphere. The phase of the as-cast alloys were characterized by X-ray diffraction (XRD) using Cu K α radiation (PANalytical B.V./X'Pert Pro diffractometer). Samples for observation were cut from the alloy rods, mechanically ground and polished through standard routines. The microstructure and chemical composition were examined on the polished samples using a scanning electron microscopy (SEM, SSX-550) equipped with energy dispersive spectrometer (EDS). Hardness was characterized using a Vickers microhardness tester (Wolpert 402MVD) under a load of 200 g for 20 s. The hardness measurements were made on at least seven points to yield an average value for each sample. Compression tests were performed on a CMT5105 machine using a strain rate of $2/3 \times 10^{-4} \text{ s}^{-1}$, where the test samples were 5 mm in diameter and 10 mm in length.

3. Results

Fig. 1 presents the XRD patterns of the FeCoNiCuAl quinary alloy and FeCoNiCuAlX alloys. The Bragg peaks in the XRD pattern of the FeCoNiCuAl quinary alloy can be indexed to simple FCC and BCC phases. The much higher peak intensity of the BCC phase suggests that the BCC structure is a dominative phase in the alloy. The addition of Si, Cr or Ti element into the FeCoNiCuAl alloy has no obvious effect on the phase constituents in these alloys as

shown in Fig. 1(b)–(d). All the three alloys consist of a dominated BCC phase and a small amount of FCC phase. However, the peak positions of the FCC and BCC phases varied with the added element. Fig. 2 displays the Bragg peaks of the FeCoNiCuAl, FeCoNiCuAlSi, FeCoNiCuAlCr and FeCoNiCuAlTi alloys in the range of $2\theta = 40\text{--}48^\circ$. It is clear that the addition of Si moves the Bragg peak (110) of the BCC phase to a larger angle, while shifts the peak (111) of the FCC phase to a smaller angle. The element Cr has no obvious effect on the positions of the Bragg peaks of the FCC and BCC phases. And the element Ti shifts the Bragg peak (110) of the BCC phase to a smaller angle. These results suggest that the addition of Si to the FeCoNiCuAl alloys reduces the lattice constant of BCC phase in the alloy, Cr has no obvious effect on the lattice constant of the BCC phase, and the addition of Ti increases the lattice constant of the BCC phase. Table 1 lists the lattice constant of the FCC and BCC phases in these alloys, which are calculated from the XRD data. It clearly shows that the additions of Si and Ti affect the lattice constant of the BCC and FCC phases. A few sharp Bragg peaks in Fig. 1(e) could be indexed to the FCC, BCC and intermetallic compound ZrFe_3Al phases, suggesting that the addition of Zr into the FeCoNiCuAl alloy retards the formation of the solid solution, and enhances the formation of the intermetallic compounds. Many sharp Bragg peaks, which have been found on the XRD pattern of the FeCoNiCuAlNd alloy as shown in Fig. 1(f), cannot be indexed to the simple FCC and BCC structure, meaning that the complex compounds are formed in the FeCoNiCuAlNd alloy. The compounds formed in the FeCoNiCuAlNd alloy cannot be exactly identified from the XRD pattern due to its complexity. From the results above, we conclude that the simple FCC and BCC solid solutions are formed in the FeCoNiCuAl, FeCoNiCuAlSi, FeCoNiCuAlCr and FeCoNiCuAlTi alloys, while the Zr and Nd retards the formation of the solid solutions.

Fig. 3 gives the back-scattering electron images of the as-cast FeCoNiCuAl and FeCoNiCuAlX alloys. The alloys excluding the FeCoNiCuAlZr exhibit a typical cast dendrite microstructure with different morphologies of dendrite (DR) and interdendrite (ID), while the FeCoNiCuAlZr alloy displays a microstructure with a white plate-like structure and smaller black region (marked as A and B in Fig. 3(e), respectively). The obvious elemental segregations are clearly observed in all the alloys. Table 2 lists the distributions of elements in the different regions marked in Fig. 3. It has been found that the Cu element is enriched in the ID region in the FeCoNiCuAl, FeCoNiCuAlSi, FeCoNiCuAlCr and FeCoNiCuAlTi alloys. The addition of Si, Cr or Ti element enhances the Cu segregation in the ID regions, and the added element (Si, Cr or Ti) segregates in the DR region of the alloys.

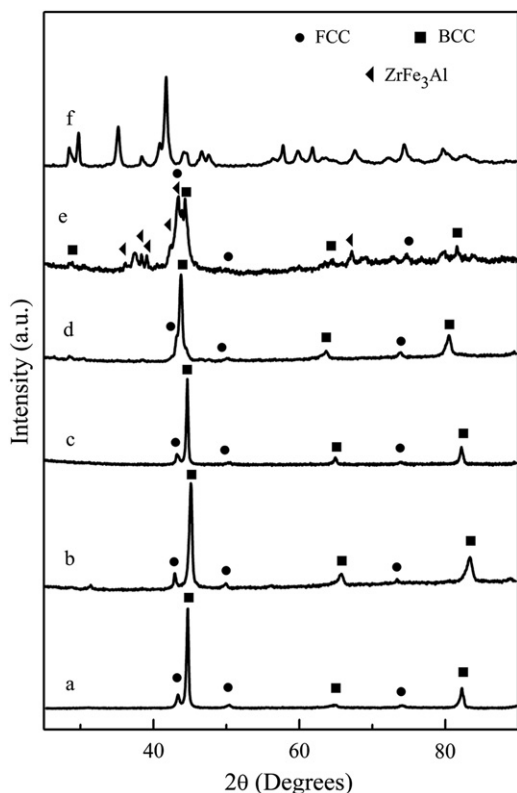


Fig. 1. XRD patterns of the as-cast alloys. (a) FeCoNiCuAl; (b) FeCoNiCuAlSi; (c) FeCoNiCuAlCr; (d) FeCoNiCuAlTi; (e) FeCoNiCuAlZr; and (f) FeCoNiCuAlNd.

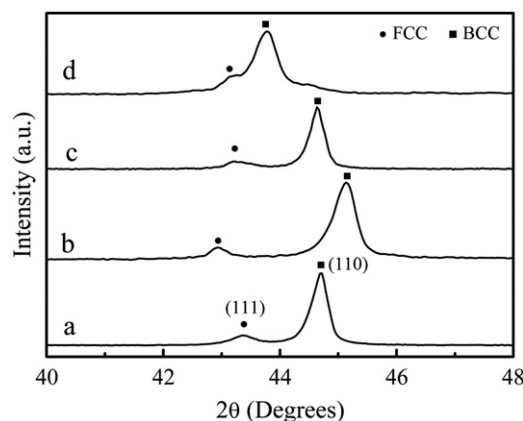


Fig. 2. XRD patterns of the as-cast alloys. (a) FeCoNiCuAl; (b) FeCoNiCuAlSi; (c) FeCoNiCuAlCr; and (d) FeCoNiCuAlTi.

Combining with XRD patterns of the four alloys, we can conclude that these alloys have a dominated BCC dendrite phase and a small amount of FCC interdendrite phase with Cu enrichment. It should be noted that the finer microstructure can be found in the dendrite of the FeCoNiCuAlTi alloy, and the grey region (marked as A in Fig. 3(d)), which has higher Fe content, exists around the core of the dendrite (DR region). However, the third phase has not been detected in the XRD pattern of the alloy, which might indicate that the phase has similar lattice constant and structure with the other phases. The FeCoNiCuAlZr alloy has different microstructure with that of the FeCoNiCuAl alloy. It can be seen from the back-scattering electron image that the FeCoNiCuAlZr alloy mainly consists of a Zr-enriched region and a Zr-depleted region (marked as A and B in Fig. 3(e), respectively). When the rare earth Nd is added into the FeCoNiCuAl alloy, the typical cast

dendrite microstructure remains unchanged. However, the finer structure can be found. The core of the dendrite, i.e. the α phase is enriched with (Fe, Co) and depleted with (Nd, Cu); the interdendrite phase, i.e. the γ phase is enriched with Nd and Cu elements, and the β phase has a composition between them. It could be deduced that the (Fe, Co)-enriched phase will solidify first.

Table 1

Lattice constant of the BCC and FCC phases in the as-prepared alloys.

Compositions	Lattice constant of BCC phase (Å)	Lattice constant of FCC phase (Å)
FeCoNiCuAl	2.8670	3.6156
FeCoNiCuAlSi	2.8357	3.6465
FeCoNiCuAlCr	2.8711	3.6136
FeCoNiCuAlTi	2.9110	3.6304

Table 2

Distributions of elements in different regions of the alloys (at%).

Alloys	Regions in Fig. 3	Fe	Co	Ni	Cu	Al	X
FeCoNiCuAl	DR	22.5	23.0	19.7	15.2	19.6	0
	ID	16.5	15.8	18.4	35.9	13.4	0
FeCoNiCuAlSi	DR	16.9	19.9	17.2	11.8	18.5	15.7
	ID	4.7	6.0	4.5	68.3	14.0	2.5
FeCoNiCuAlCr	DR	21.3	21.8	19.4	7.7	11.6	18.2
	ID	9.6	10.6	20.3	43.7	9.0	6.8
FeCoNiCuAlTi	DR	12.7	23.0	15.6	2.9	25.8	20.0
	ID	8.3	5.3	7.9	67.1	8.7	2.7
	A	33.4	17.2	19.1	8.2	8.4	13.7
FeCoNiCuAlZr	A	18.8	19.2	17.3	9.2	12.2	23.3
	B	15.0	17.7	18.1	12.8	33.7	2.7
FeCoNiCuAlNd	α	46.9	27.1	8.3	0.9	16.7	0.1
	β	11.7	27.0	16.5	15.0	15.0	14.8
	γ	1.8	6.0	17.6	31.3	7.1	36.2

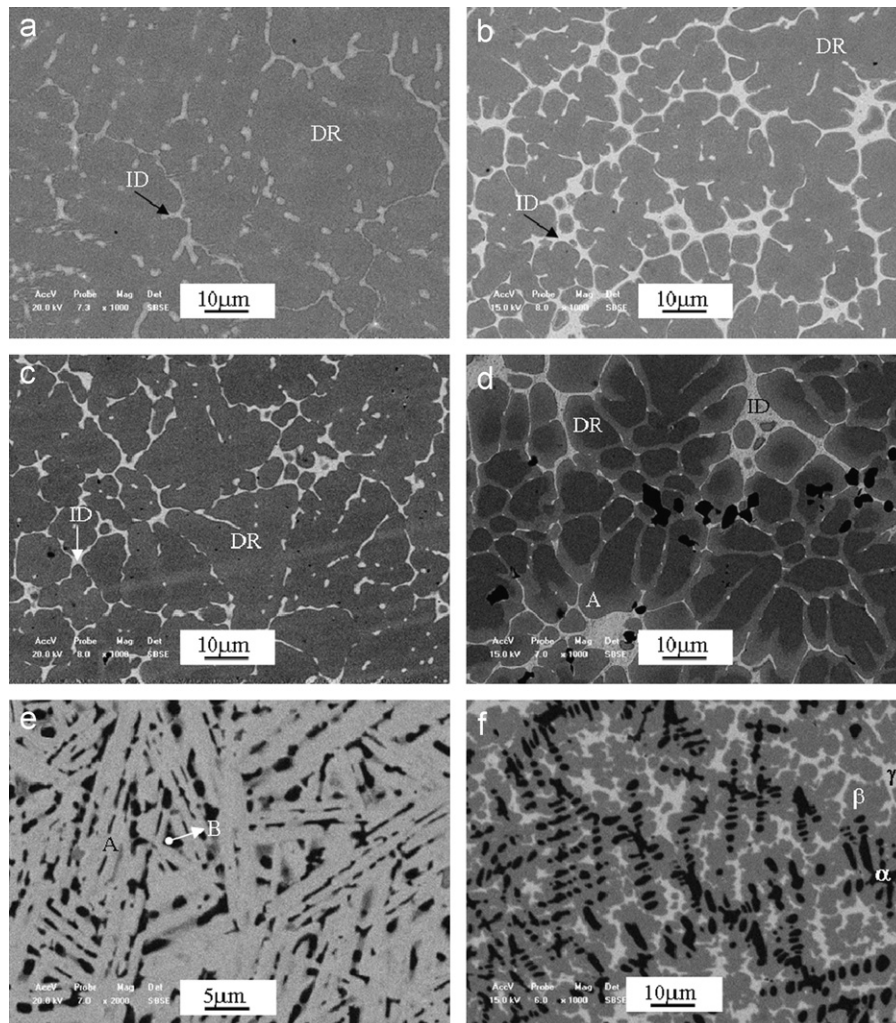


Fig. 3. Back scattering SEM images of the as-cast alloys. (a) FeCoNiCuAl; (b) FeCoNiCuAlSi; (c) FeCoNiCuAlCr; (d) FeCoNiCuAlTi; (e) FeCoNiCuAlZr; and (f) FeCoNiCuAlNd.

The mixing entropy of Fe–Nd and Fe–Cu are 1 and 15 kJ/mol, thus the Nd and Cu will remain in the residual liquid phase, and finally go to the latest solidified interdendrite γ phase.

Fig. 4 summarizes the room temperature compressive true stress–strain curves of all the six alloys. The FeCoNiCuAl alloy has the maximum compressive strength of 1.33 GPa and the total true strain of 9%. The addition of Si element increases the maximum compressive strength up to 1.93 GPa and the total true strain to 10%; the FeCoNiCuAlTi alloy has the maximum compressive strength of 1.66 GPa and the total true strain of 11.0%; and the FeCoNiCuAlCr alloy has the maximum compressive strength of 1.51 GPa and the total true strain of 16.0%. It is clear that addition of Si, Ti or Cr element could enhance both the compressive strength and the plasticity of the alloys. However, the Zr and Nd elements retard the compressive properties of the alloys. The FeCoNiCuAlZr and FeCoNiCuAlNd alloys have the maximum compressive strength of 1.05 GPa and 0.67 GPa, respectively. Both of the alloys display a brittle fracture without any plasticity observed, which could be attributed to the formation of brittle intermetallic compounds formed in the two alloys. Fig. 5 gives the hardness values of the six alloys. The FeCoNiCuAlSi alloy has the highest hardness value of 680.4 HV, and the FeCoNiCuAlCr and FeCoNiCuAlZr have the lowest hardness of 472.0 HV. The other three alloys have hardness values between them, i.e. the FeCoNiCuAl, the FeCoNiCuAlTi and the FeCoNiCuAlNd alloys have the hardness of 536.2 HV, 625.3 HV and 577.6 HV, respectively.

4. Discussions

The multi-component alloys have high mixing entropy. The mixing entropy of an alloy with 5 or 6 elements in equal-molar ratio is 13.37 and 14.90 J/kmol, respectively. The high mixing

entropy can effectively increase the extent of confusion in the alloys, and is benefit to the formation of a solid solution. However, the addition of some elements into the FeCoNiCuAl alloy retards the formation of solid-solution, indicating that the high mixing entropy is not the only factor to determine the formation of the solid solution in a multi-elements alloy. Different effects of Fe, Ag and Au on the microstructure, phase constituents and properties have also been observed in the 6-elements AlCoCrCuNi-based alloys [16]; while the addition of Mn, Ti and V into the AlCoFeCoNiCu alloy induces different response in the microstructure, phases and properties of the equal-molar alloys [25]. All the results suggest that other factor but the mixing entropy exists in determining the formation of a solid-solution. Zhang et al. related the formation of the solid solution to the mixing enthalpy (ΔH_{mix}) and atomic size difference (δ), and found that the small atomic size differences and near-zero values of the mixing enthalpy are responsible for the formation of a disorder solid solution [8]. Liu et al. further confirm that the mixing enthalpy determines whether the solid solution phases or compounds form in the nearly equiatomic multi-principal alloy systems, and the valence electron concentration is a key parameter to control the phase stability for the FCC or BCC solid solutions [21].

Following the Hume–Rothery rule, crystal structure, atomic size difference, electronegativity difference ($\Delta\chi$), and electron concentration of the constituent elements are key factors in determining the formation of the solid solution. We have calculated the values of ΔH_{mix} , δ , $\Delta\chi$ and valence electron concentration (VEC) which is used to represent the electron concentration effect. The δ , ΔH_{mix} , $\Delta\chi$ and VEC in an alloy system are defined as [8,21]

$$\delta = 100\sqrt{\sum_{i=1}^N C_i(1-r_i/\bar{r})^2}$$
 (1)

$$\Delta H_{mix} = \sum_{i=1, i \neq j}^N \Omega_{ij} C_i C_j$$
 (2)

$$\Delta\chi = \sqrt{\sum_{i=1}^N C_i(\chi_i - \bar{\chi})^2}$$
 (3)

$$VEC = \sum_{i=1}^N C_i(VEC)_i$$
 (4)

where N is the number of the components in an alloy system, C_i and C_j are the atomic percentages of the i th and j th element, r_i is the atomic radius of i th element which can be obtained from Ref. [23], $\bar{r} = \sum_{i=1}^N C_i r_i$ is the average atomic radius, $\Omega_{ij} = 4\Delta H_{AB}^{mix}$ is the regular melt-interaction parameter between i th and j th elements, ΔH_{AB}^{mix} is the mixing enthalpy of binary AB alloys which can be found in Ref. [24], χ_i is the Pauling electronegativity for the i th element, $\bar{\chi} = \sum_{i=1}^N C_i \chi_i$ is the average electronegativity, and $(VEC)_i$ is the VEC for the i th element. Table 3 gives the δ , ΔH_{mix} , $\Delta\chi$ and VEC for the six alloys investigated. The relationship between δ and ΔH_{mix} of all the alloys except the FeCoNiCuAlSi alloy is in agreement with solid-solution formation rules for the multi-component high-entropy alloys proposed by Zhang et al. [8], where solid solution or order solid solution (i.e. minor solid solution precipitation beside

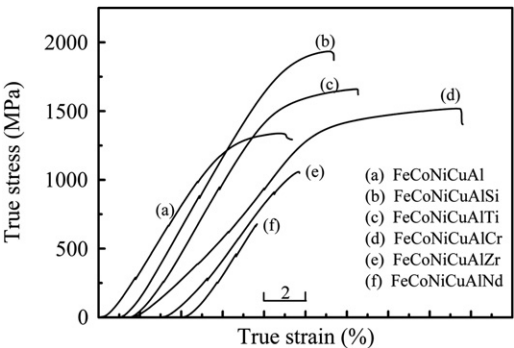


Fig. 4. Room temperature compressive true stress–strain curves of the as-cast FeCoNiCuAlX alloys.

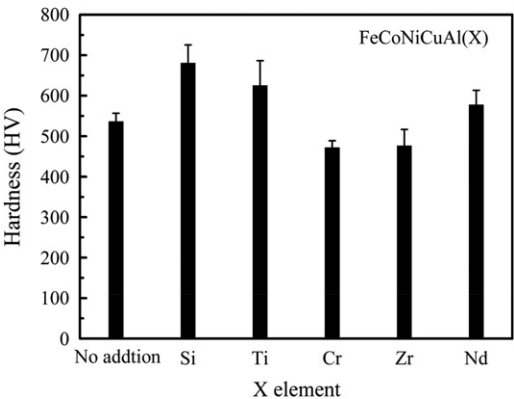


Fig. 5. Hardness values of the as-cast FeCoNiCuAlX alloys.

Table 3
Values of δ , ΔH_{mix} , $\Delta\chi$ and VEC of various multi-principal-element alloys.

Alloys	ΔH_{mix} (kJ/mol)	δ	$\Delta\chi$	VEC	Structures
FeCoNiCuAl	−5.28	5.24	0.154	8.2	FCC+BCC
FeCoNiCuAlSi	−20.45	4.83	0.141	7.50	FCC+BCC
FeCoNiCuAlCr	−4.78	4.83	0.151	7.84	FCC+BCC
FeCoNiCuAlTi	−16.90	6.58	0.172	7.50	FCC+BCC
FeCoNiCuAlZr	−23.90	9.59	0.226	7.50	Compounds
FeCoNiCuAlNd	−15.78	14.82	0.285	7.83	Compounds

Table 4

Atomic radii of different elements [23] and mixing enthalpies of atomic pairs (kJ/mol) [24].

Fe 1.27 Å	Co 1.25 Å	Ni 1.25 Å	Cu 1.28 Å	Al 1.43 Å	Si 1.32 Å	Cr 1.28 Å	Ti 1.46 Å	Zr 1.60 Å	Nd 1.82 Å
Fe	–1	–2	13	–11	–35	–1	–17	–25	1
	Co	0	6	–19	–38	–4	–28	–41	–20
		Ni	4	–22	–40	–7	–35	–49	–30
			Cu	–1	–19	12	–9	–23	–22
				Al	–19	–10	–30	–44	–38

solid solution) are formed when $\delta < 6.25$ and ΔH_{mix} lies between -20 and 5 kJ/mol. It can also be found that the value of VEC has no clear difference between the solid-solution-forming alloys and the non-solid-solution-forming alloys. On the other hand, all the solid-solution forming alloys investigated in the paper have both smaller δ and smaller $\Delta\chi$, while the compound-forming or amorphous/compound-forming alloys have larger δ and larger $\Delta\chi$. The phenomenon indicates that the $\Delta\chi$ is another possible parameter to estimate whether the solid solution could be formed in an alloy. In fact, the result is not surprising since both the mixing enthalpy and the electronegativity difference are related to the chemical compatibility.

Table 4 lists the atomic radii of the constituent elements and the mixing enthalpies of the atomic pairs involved. The positive mixing enthalpies between Cu and the other elements in the FeCoNiCuAl alloy lead to the repulsion of the Cu atoms. Therefore, the Cu atoms are rejected to the grain boundary during solidification, and form Cu-rich solid solution in the FeCoNiCuAl alloy. The phenomenon has also been observed in other systems [2,19,20,22]. Though the mixing enthalpies between Si and Cu are -19 kJ/mol, it is still a relative positive value among the mixing enthalpies between Si and the other elements. Therefore, the addition of Si further rejects the Cu atoms into the grain boundary, and enhances the Cu segregation in the interdendrite FCC phase. The similar situation exists in the FeCoNiCuAlCr and FeCoNiCuAlTi alloys, which explains the increased segregation of Cu in ID regions with the addition of Si, Cr and Ti.

The average atomic radius of FeCoNiCuAl alloy is 1.296 Å. The atomic radius of Cr is 1.28 Å, which is close to the value of the average atomic radius of the FeCoNiCuAl. Therefore, not only the lattice constant of the BCC and FCC phases (refer to Table 1) but also the compressive strength remains nearly unchanged with the addition of Cr. The addition of Si or Ti, whose atomic radius are 1.32 and 1.46 Å, has no obvious effect on phase constituents of the alloys, however they cause the distortion of the lattice, and increase the lattice distortion energy. The solid solution strengthening mechanisms of Si and Ti atoms could be one of the reasons to explain the improved compressive mechanical properties of the FeCoNiCuAlSi and FeCoNiCuAlTi alloys. The Zr or Nd with further larger atomic size destroys the formation of the solid solutions, and retards the mechanical properties of the alloys.

5. Conclusions

The FeCoNiCuAl and FeCoNiCuAlX (X refers to Si, Cr, Ti, Zr, or Nd) alloys were prepared using the suck-casting into a copper-mold method. The FeCoNiCuAl, FeCoNiCuAlSi, FeCoNiCuAlCr, and FeCoNiCuAlTi alloys mainly consist of simple FCC+BCC solid solution phases, where the BCC is a dominated Cu-depleted phase. The Si, Cr or Ti segregates in the dendrite region of the alloy, and enhances the Cu segregation in the interdendrite regions. On the other hand,

the addition of Zr or Nd into the FeCoNiCuAl alloy eliminates the formation of the solid solution phases, and form complex inter-metallic compounds in the alloys. The smaller atomic size difference and smaller electronegativity difference of the constituent elements are benefit to the formation of the solid solution. The compressive strength and plasticity of the alloys are enhanced by the addition of Si and Ti, and retarded by the addition of Zr or Nd.

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